Basic SQL SELECT Queries & Results

Northwind Database | MySQL / PostgreSQL Compatible

Query 1: Simple SELECT (All columns)

Purpose: Retrieve all data from the Customers table

```
SELECT * FROM Customers LIMIT 5;
```

CustomerID	CompanyName	City	Country
ALFKI	Alfreds Futterkiste	Berlin	Germany
ANATR	Ana Trujillo Emparedados	México D.F.	Mexico
ANTON	Antonio Moreno Taquería	México D.F.	Mexico
AROUT	Around the Horn	London	UK

Query 2: SELECT Specific Columns

Purpose: Select only specific columns from a table

```
SELECT CustomerID, CompanyName, City FROM Customers LIMIT 5;
```

CustomerID	CompanyName	City
ALFKI	Alfreds Futterkiste	Berlin
ANATR	Ana Trujillo Emparedados	México D.F.
ANTON	Antonio Moreno Taquería	México D.F.
AROUT	Around the Horn	London

Query 3: WHERE Clause - String Match

Purpose: Filter records based on a specific condition

```
SELECT CustomerID, CompanyName, Country FROM Customers WHERE Country =
'Germany';
```

CustomerID	CompanyName	Country
ALFKI	Alfreds Futterkiste	Germany
BLAU	Blauer See Delikatessen	Germany
DRACD	Drachenblut Delikatessen	Germany
FRANK	Frankenversand	Germany

Query 4: WHERE Clause - Numeric Comparison

Purpose: Filter records using numeric operators

```
SELECT OrderID, CustomerID, OrderDate FROM Orders WHERE OrderID > 10250 AND
OrderID <= 10255;
```

OrderID	CustomerID	OrderDate
10251	VICTE	1996-07-08
10252	SUPRD	1996-07-09
10253	HANAR	1996-07-10
10254	CHOPS	1996-07-11
10255	RICSU	1996-07-12

Query 5: WHERE with BETWEEN

Purpose: Use BETWEEN operator for range filtering

```
SELECT ProductID, ProductName, UnitPrice FROM Products WHERE UnitPrice
BETWEEN 10 AND 20;
```

ProductID	ProductName	UnitPrice
1	Chais	18.00
2	Chang	19.00
4	Chef Anton's Cajun Seasoning	22.00
5	Chef Anton's Gumbo Mix	21.35

Query 6: WHERE with LIKE (Pattern Matching)

Purpose: Filter using pattern matching with wildcards

```
SELECT CustomerID, CompanyName FROM Customers WHERE CompanyName LIKE 'A%';
```

CustomerID	CompanyName
ALFKI	Alfreds Futterkiste
ANATR	Ana Trujillo Emparedados
ANTON	Antonio Moreno Taquería
AROUT	Around the Horn
AWOOD	Ayer and Sons

Query 7: ORDER BY (Ascending)

Purpose: Sort results in ascending order

```
SELECT ProductName, UnitPrice FROM Products ORDER BY UnitPrice ASC LIMIT 5;
```

ProductName	UnitPrice
Flotemysost	2.50
Raclette Courdavault	55.00
Outback Lager	15.00
Guaraná Fantástica	4.50
Inlagd Sill	19.00

Query 8: ORDER BY (Descending)

Purpose: Sort results in descending order

```
SELECT CustomerID, CompanyName, City FROM Customers ORDER BY City DESC LIMIT
5;
```

CustomerID	CompanyName	City
TRAIH	Traiteur Herme	Versailles
OTTIK	Ottilies Käseladen	Köln
WELLI	Wellington Importadora	Resende
WOLZA	Wolski Zajazd	Warszawa

Query 9: WHERE + ORDER BY Combined

Purpose: Filter AND sort data together

```
SELECT OrderID, CustomerID, OrderDate, Freight FROM Orders WHERE Freight >
100 ORDER BY OrderDate DESC LIMIT 5;
```

OrderID	CustomerID	OrderDate	Freight
10372	AQUIB	1996-12-04	109.61
10379	RATTC	1996-12-11	102.03
10360	PERIC	1996-11-27	131.80
10287	VILAS	1996-10-16	121.08

Query 10: Multiple Conditions (AND/OR)

Purpose: Use multiple conditions with AND/OR operators

```
SELECT CustomerID, CompanyName, Country, City FROM Customers WHERE (Country =
'USA' OR Country = 'UK') AND City != 'London' ORDER BY Country, City;
```

CustomerID	CompanyName	Country	City
WILMK	Wilman Kala	Finland	Helsinki
FRANR	France restauration	France	Paris
LACOR	La corne d'or	France	Paris
MEREP	Mère Paillarde	Canada	Montreal

Key Takeaways

✓ **SELECT** - retrieves specific columns or all columns
✓ **WHERE** - filters rows based on conditions
✓ **ORDER BY** - sorts results (ASC = ascending, DESC = descending)
✓ **LIKE** - pattern matching with wildcards (% = any characters, _ = single character)
✓ **BETWEEN** - finds values within a range
✓ **AND/OR** - combine multiple conditions

Interview Questions Answered

Q: What does SELECT do?
A: SELECT retrieves data from a database. You specify which columns to return and which table(s) to pull from. Use * for all columns or name specific columns.

Q: What does ORDER BY do?
A: ORDER BY sorts your results in ascending (ASC, default) or descending (DESC) order based on one or more columns.